

The square-root interior endpoint profile

Complete theorem; independently reviewed

30 September 2026

Theorem 1. *Suppose $p_1, \dots, p_m$ are nondecreasing $[0, 1]$ -valued columns on a finite ordered set with positive probability row weights w , and*

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Put $u_i = 1 - p_i$ and $s = \sum_i u_i$. For every fixed $0 < \delta < L < \infty$, uniformly over rows with $\delta \leq s \leq L$,

$$\max_i |mu_i - s| \leq C_{\delta, L} m^{-1/2}.$$

We use the exact marked identity and notation from `marked_cancellation_identity.tex`, and the common-coordinate bookkeeping from `sharp_interior_endpoint_profile.tex`. The present argument replaces its infinite Laguerre kernel by a positive finite polynomial. We give all changes to the measure and maximum arguments explicitly; Gaussian tails are not assumed here.

The new analytic inputs are `../rational_designs/asymptotic/finite_jackson_endpoint_kernel.tex` (including its global low-order off-annulus estimate) and `../bernstein_walsh_polarization_tail.tex`. As a prior estimate we use any independently proved bound

$$\max_i |mu_i - s| \leq R_m \leq m^{-\alpha}, \quad \alpha > 0, \tag{1}$$

on each fixed bounded s -interval. The earlier polynomial-rate profile theorem suffices; the stronger logarithmic square-root theorem is not needed as an input.

Parameters and local comparisons. Fix a large error exponent $A > 10$. Fix the positive integer r in the Jackson kernel sufficiently large that

$$2r\alpha > A + 10, \quad r > A + 10. \tag{2}$$

The polynomial exponents absorbed here are fixed independently of r . Constants, as distinct from powers of m , may depend on r . Next choose the degree ratio sufficiently small for both the global Bernstein–Walsh theorem and the Jackson off-annulus geometric sum (the latter constants depend on r), and finally choose its differential cutoff $J = B \log m$. For the Jackson kernel $Q_m(x, a)$ let

$$b = \sqrt{\ell}/\nu \asymp m^{-1/2}, \quad E_b(x, a) = b^{-1}(1 + |x - a|/b)^{-2r}.$$

Here $\ell = \lfloor (m - 1)/4 \rfloor$, and the integer $\nu \asymp m$ has fixed sufficiently small ratio, after r is fixed.

For marked column i , put $M = m - 1$, $N = M - \ell$, $z_j = \ell u_j$, and use

$$x_i = M^{-1} \sum_{j \neq i} z_j, \quad h_i = z_i - x_i, \quad d\mu_i = (\ell + 1) \mathbb{E}_B \prod_{j \in B} p_j dw,$$

where B ranges uniformly over the ℓ -subsets of the other M indices. Each μ_i is a probability measure. The reference measure has density

$$d\beta_\ell(x) = \frac{\ell+1}{\ell}(1-x/\ell)^\ell \mathbf{1}_{[0,\ell]}(x) dx.$$

Let $H = \max_j z_j - \min_j z_j$.

The exact polynomial identities, global high-order tail theorem, Jackson off-annulus estimate, and compact subset comparisons give, for centers in any fixed positive compact interval,

$$|\mu_i(Q_m(x_i, a)) - \beta_\ell(Q_m(\cdot, a))| \leq \frac{C}{mb^2} \int_{x_i \in W} H^2 E_b(x_i, a) d\mu_i + m^{-A}, \quad (3)$$

$$|\mu_i(h_i Q_m(x_i, a))| \leq \frac{C}{m} \int_{x_i \in W} H^2 (1 + b^{-1} + Hb^{-2}) E_b(x_i, a) d\mu_i + m^{-A}. \quad (4)$$

Here W is a sufficiently large fixed positive row annulus, and constants or the envelope scale can change by fixed factors. To see the order of these operations, first apply the exact marked and unmarked identities to the polynomial Q_m , whose degree is at most a small fixed multiple of m . Remove differential orders above J by the global bounded-polynomial tail theorem. At orders up to J , discard rows outside W using the Jackson angular localization theorem. On W use the finite centered differential sum and the success-weighted subset Taylor comparison. The prior estimate makes H small there, and finite balance ensures every marked value is bounded whenever x_i is bounded. The scalar terms outside W are also harmless: the global scalar kernel tail is $O(b^{2r-1})$, each row factor is polynomially bounded, and (2) absorbs their fixed powers of m .

Mass absorption with polynomial tails. We record why the earlier Gaussian mass argument still applies. On W , (1) gives $H \leq CR_m$, so (3) has right side at most

$$\epsilon_m \mu_i(E_b(x_i, a)) + m^{-A}, \quad \epsilon_m = CR_m^2/(mb^2) \rightarrow 0.$$

The Jackson kernel is at least c/b on $|x - a| \leq c_r b$, and its reference integral is bounded above and below by positive constants. Fix an outer positive annulus I and let $\phi(a) = \text{dist}(a, \partial I)^2$. Maximize $\phi(a) \mu_i([a - c_r b, a + c_r b])/b$ over $a \in I$. If this supremum is at least one, the total-mass bound implies that an approximate maximizer has boundary distance $d \geq c\sqrt{b}$. On its $d/2$ neighborhood the weights ϕ are comparable, and summing the envelope over intervals of length $c_r b$ bounds the near part by a constant times the supremum divided by $\phi(a)$. The far part is at most

$$Cb^{-1}(d/b)^{-2r} = O(b^{r-1}),$$

which tends to zero. Thus the same weighted-supremum absorption proves a uniform local interval bound. On any strictly smaller annulus, it yields

$$\mu_i(|x_i - a| \leq b) \leq Cb, \quad c \leq \mu_i(Q_m(x_i, a)) \leq C, \quad \mu_i(E_b(x_i, a)) \leq C. \quad (5)$$

Summation over length- b intervals also gives, for $T \geq 1$,

$$\int_{|x_i - a| \geq Tb} E_b(x_i, a) d\mu_i \leq CT^{1-2r} + O(m^{-A}), \quad (6)$$

as long as the local interval argument is applied in a fixed larger annulus. Tails beyond that larger annulus are bounded directly by the total mass and the global scalar kernel bound. By choosing r as in (2), those tails have any required fixed inverse-power accuracy in this proof.

The weighted maximum and its far tail. Use the common coordinate and deviations

$$y = \ell s/m, \quad h_i^* = z_i - y, \quad x_i = y - h_i^*/M, \quad h_i = (m/M)h_i^*.$$

Choose a fixed positive annulus $I = [a_-, a_+]$ whose strictly smaller interior contains the desired y -range. The mass bounds are first proved on an even larger annulus. Let

$$D = \max_{i, y \in I} \text{dist}(y, \partial I) |h_i^*(y)|.$$

If $D > C_* b$, select a maximizing row y_0 and column i , and write $\Delta = |h_i^*(y_0)|$, $d_0 = \text{dist}(y_0, \partial I)$. Then

$$d_0/b = D/(b\Delta) \geq C_*/R_m \geq C_* m^\alpha, \quad \Delta \geq c_I C_* b.$$

In the $d_0/2$ row neighborhood, weighted maximality gives $|h_j^*| \leq C\Delta$ for every j , hence $H \leq C\Delta$. The deleted coordinate differs from y by $O(R_m/m) = o(b)$. Any row entering the fixed deleted-coordinate neighborhood has bounded full s by finite balance, so this estimate applies before restricting to the common-coordinate neighborhood.

First discard rows outside a fixed positive enlargement by the global Jackson off-annulus and scalar tail estimates already used in (3)–(4). The scalar Jackson tail there is $O(b^{2r-1}(1+x)^{-r})$; we do not bound it by E_b globally. Within that fixed enlargement, $Q_m \leq CE_b$, and every far contribution beyond a fixed multiple of d_0 has envelope at most

$$Cb^{-1}(d_0/b)^{-2r} \leq Cm^{1/2-2r\alpha}.$$

The measures have mass one and all scalar prefactors used here are bounded by Cm^4 (in fact much smaller bounds suffice). Consequently (2) makes all those contributions smaller than m^{-A} . This bound has an exponent independent of r except for the favorable decay term $-2r\alpha$.

The sign test. If the selected deviation is positive, set $a = x_i(y_0) + A_0 b$; for a negative deviation use the minus sign. Choose fixed B_0 large, then $A_0 = 2B_0$. On $|x_i - a| \leq B_0 b$ the row lies strictly on the favorable side of the selected one. Monotonicity therefore gives, in the positive case,

$$h_i(x_i) \geq h_i(x_i(y_0)) - (x_i - x_i(y_0)) \geq \Delta/2,$$

provided C_* is sufficiently large. The negative case is identical with signs reversed. Strict sidedness handles flat rows. By (5)–(6), this window captures a positive amount of Q_m -mass, while the near opposite-sign contribution is at most $C\Delta B_0^{1-2r}$. Choose B_0 large enough to absorb it. The farther contribution is negligible by the preceding paragraph. Thus

$$|\mu_i(h_i Q_m(x_i, a))| \geq c\Delta.$$

But (4), the local bound $H \leq C\Delta$, and (5) give

$$|\mu_i(h_i Q_m(x_i, a))| \leq C(\Delta^2/\sqrt{m} + \Delta^3) + m^{-A} = o(\Delta),$$

because $\Delta \leq R_m \rightarrow 0$ and $\Delta \geq c_I C_* m^{-1/2}$. This contradiction proves $D = O(b)$. The cutoff is bounded below on the target range, so $h_i^* = O(m^{-1/2})$ there. Since $mu_i - s = (m/\ell)h_i^*$, Theorem 1 follows.